WEEK 10 · State-Space Estimation

The Kalman Filter and Its Extensions

Estimating unobserved levels and betas from noisy data

State space · Kalman filter · smoothing · EKF · UKF · particle filter · dynamic beta

Today's question — how do we estimate a hidden continuous state from noisy observations, and record its uncertainty along with it?

The Kalman filter is an estimator that alternates prediction and updating

Three messages that run through this lecture

1

State space separates the hidden state and the observation into two equations

The state equation describes how the hidden state moves; the observation equation, how it shows up in the data.

2

The Kalman filter is a cycle: predict → innovation → gain → update

New data enter through the innovation, the 'surprise relative to expectation', and the gain sets how much of it is used.

3

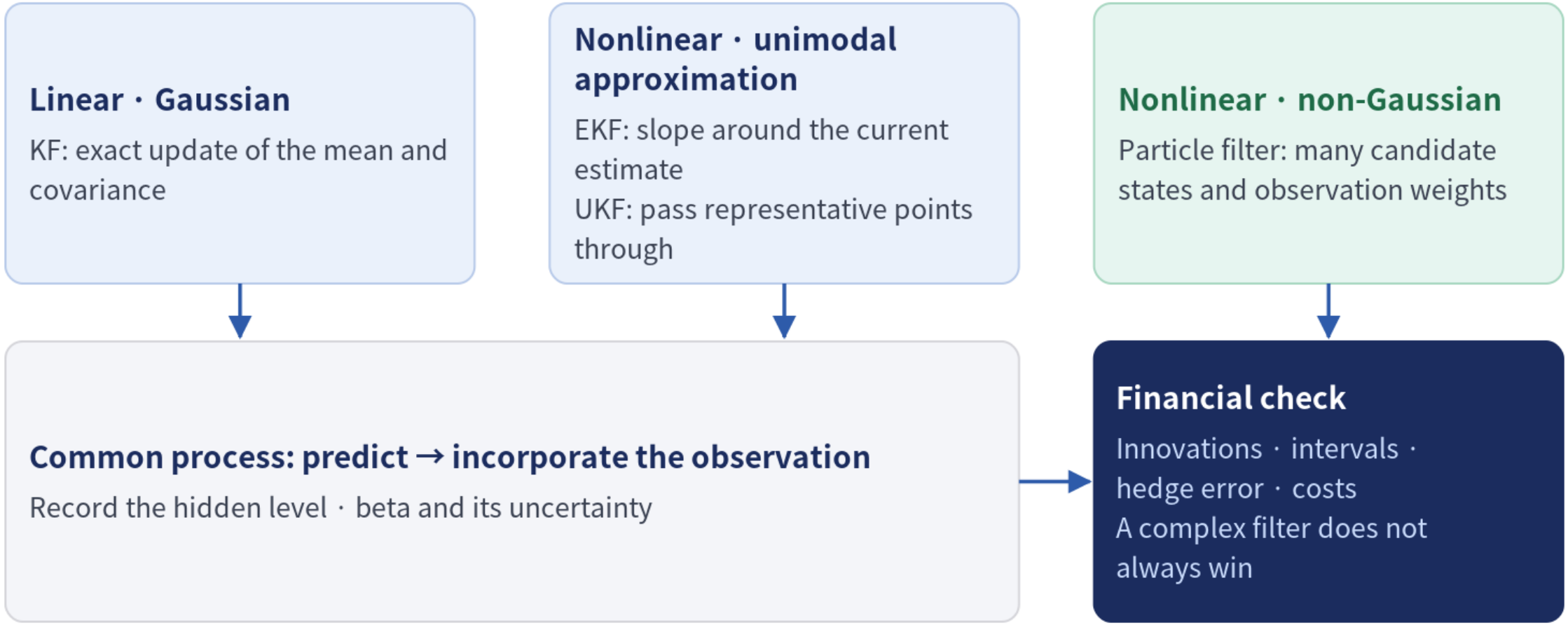
A different model calls for a different computation

EKF · UKF for nonlinearity, the particle filter for multimodal or non-Gaussian cases. The financial baseline is dynamic beta.

Filter intervals are uncertainty under the specified model and parameters — they do not cover errors in the model itself

Choose the filter by the properties of the model

KF if linear and Gaussian, EKF · UKF if nonlinear, the particle filter if multimodal — the financial evaluation is common to all



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PART 1 · Principles

State Space and the Kalman Filter

Two equations · notation · prediction and update · hand calculation

The state is hidden and observations contain noise

The state is the unobserved value the model estimates; the observation is the data actually received

Hidden state	Observation	Financial question
Latent level	Noisy price	Split level from transient trading noise?
Time-varying beta	Asset and market returns	How does the exposure change?
Common cycle factor	Production, consumption, employment	Current state of the economy? (Week 13 nowcasting)

Caution

- The latent level is a model-defined value — it does not guarantee true economic intrinsic value
- Nowcasting reads the hidden business cycle from many indicators; it is covered in detail in Week 13

State space = a state equation and an observation equation

The covariance of the state shock is Q; that of the observation noise is R

$$\underbrace{x_t = F_t x_{t-1} + w_t}$$

State equation: how the hidden state moves

$$\underbrace{y_t = H_t x_t + v_t}$$

Observation equation: how the state appears in the data

Assumptions of the basic KF

- The initial state is normal
- Each noise is normal and independent over time
- Noises are independent of each other and of the initial state

Reading the notation

- F: state transition (persistence)
- H: how the state loads on the observation
- Larger Q or R means a noisier state or observation

The time after the vertical bar is the information cutoff

The predicted and updated states are estimates of the same time-t state using different information

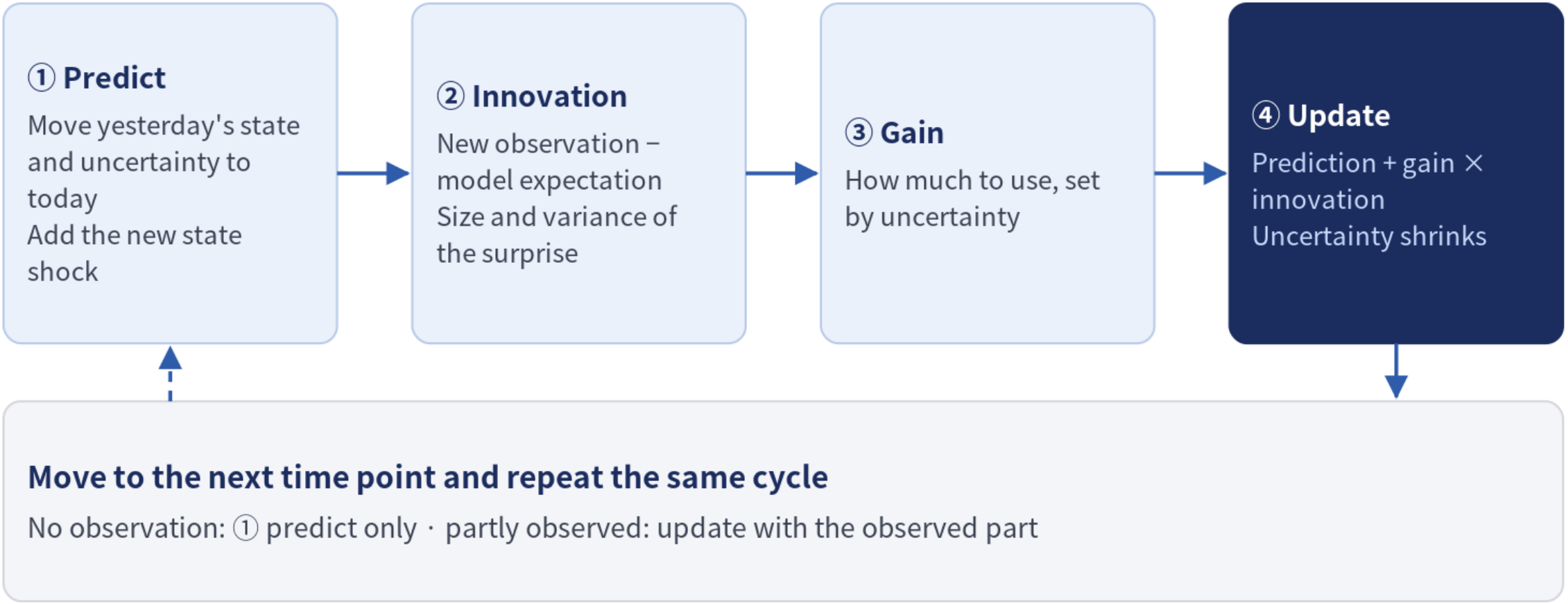
$\hat{x}_{t\|t-1}$	Today's state predicted with data up to yesterday
$\hat{x}_{t\|t}$	State updated with data up to today
P	Covariance of the state estimation error (not the variance of the observed data)

Common misconception

- P is not the variance of the observed data but the covariance of the state estimation error
- Even for today's state, the estimate before and after seeing today's data differs

The Kalman filter repeats four steps at every time point

Prediction looks ahead to today; the surprise in the observation is used in proportion to the gain



The prediction step carries yesterday's state and uncertainty forward to today

Move the mean and variance with the transition matrix F and add the variance Q of the new state shock

$$\underbrace{\hat{x}_{t|t-1} = F_t \hat{x}_{t-1|t-1}}_{\text{Predicted state}}$$

$$\underbrace{P_{t|t-1} = F_t P_{t-1|t-1} F_t^\top + Q_t}_{\text{Predicted variance}}$$

Interpretation

- When F = 1, as in the local level model, the predicted variance grows
- For a general F, the predicted variance need not exceed yesterday's

Innovation = deviation from expectation, not from yesterday

The innovation and its variance measure how surprising the new data are

$$\underbrace{e_t = y_t - H_t \hat{x}_{t|t-1}}$$

Innovation: surprise relative to the model's expectation

$$\underbrace{S_t = H_t P_{t|t-1} H_t^\top + R_t}$$

Innovation variance

Two sources of innovation variance

- Uncertainty from not knowing the state
- Observation-noise variance R

Use in diagnostics

- If the model is correct, innovations should have no autocorrelation
- If innovations keep leaning to one side, suspect the model

The gain sets how much of the innovation is used

The Kalman gain is computed from the relative size of prediction uncertainty and observation noise

$$\underbrace{K_t}_{\text{Gain}} = P_{t|t-1} H_t^\top S_t^{-1}$$

$$\boxed{\underbrace{\hat{x}_{t|t}}_{\text{Updated state}} = \hat{x}_{t|t-1} + K_t e_t}$$

$$\underbrace{P_{t|t}}_{\text{Updated variance}} = (I - K_t H_t) P_{t|t-1}$$

Interpretation

- The gain is computed from uncertainty; new information enters through the innovation

Numerical implementation

- Solve a linear system instead of inverting
- Joseph form, etc., for the covariance update

Only about 36% of the surprise in the new observation 106 is used

Local level: initial estimate 100 · initial variance 4 · state variance 1 · observation variance 9 · new observation 106

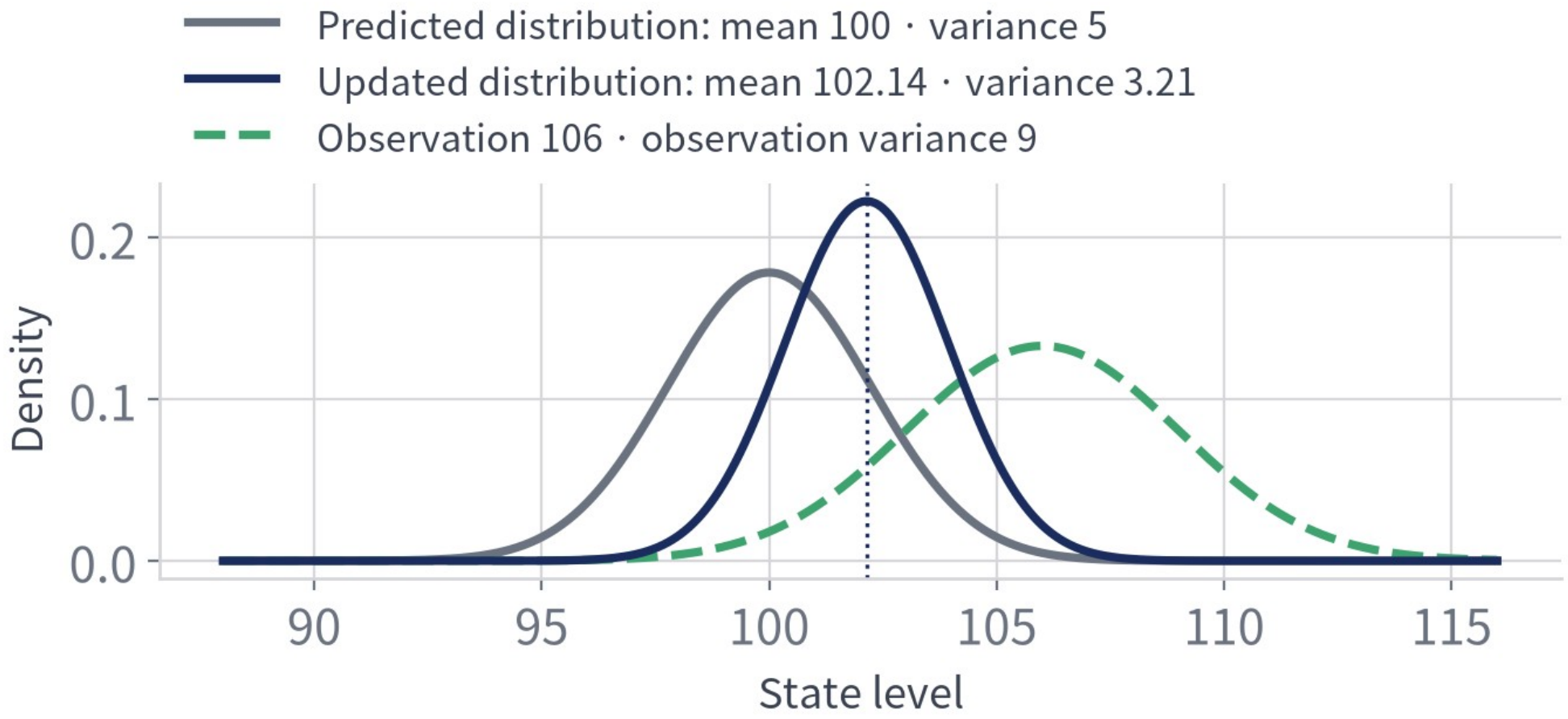
Step	Result
Predicted state	100
Predicted variance	5
Innovation	6
Innovation variance	14
Gain	≈ 0.3571
Updated state	≈ 102.1429
Updated variance	≈ 3.2143

How to read it

- 106 is not adopted as is
- Only about 35.7% of the unexpected 6 is used
- Read the gain as a 0–1 weight only in the scalar model with H = 1
- Do not read elements of a general matrix as probabilities

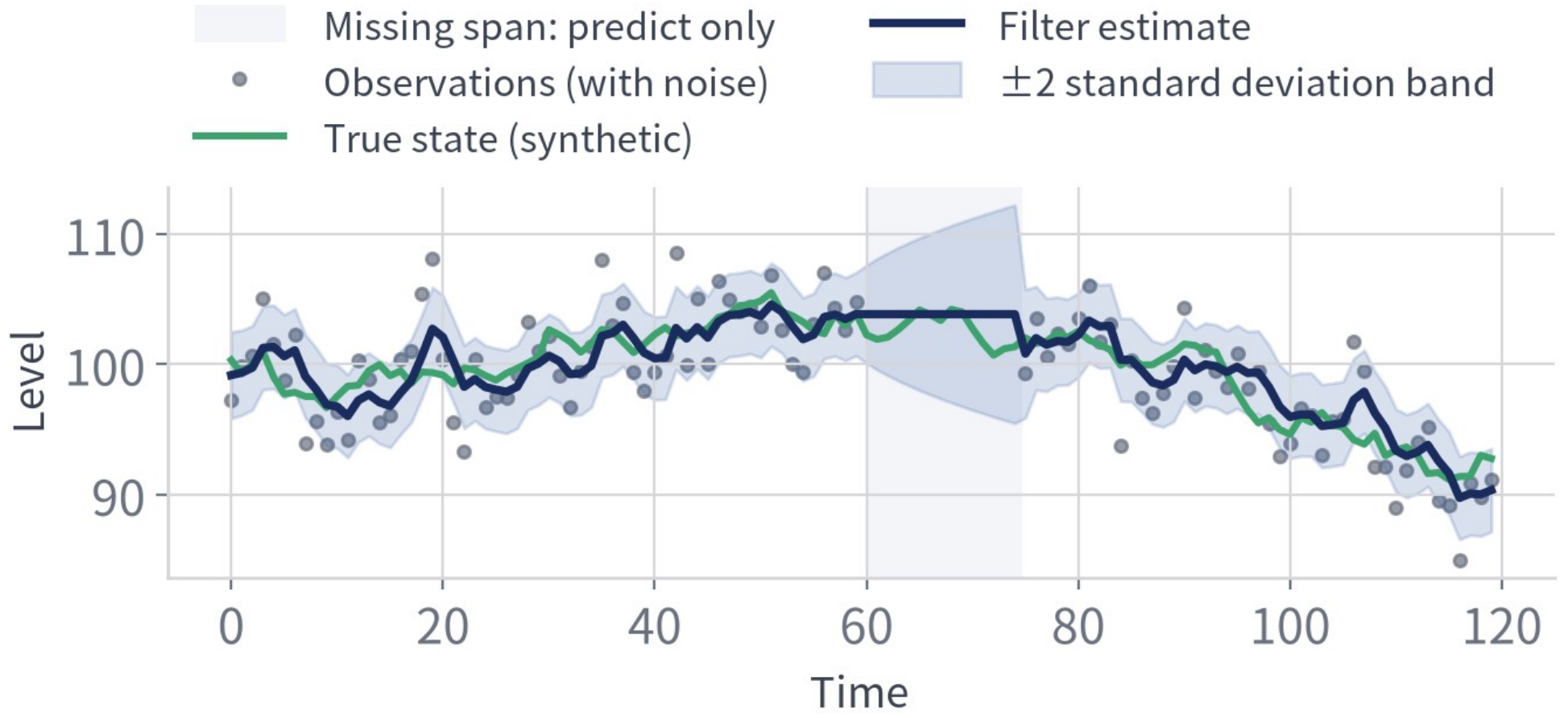
Updated distribution: between the other two, and narrower

Three distributions from the hand calculation — updated variance 3.21 is below predicted 5 and observation 9



When observations stop, only prediction runs and the interval widens

Local level synthetic data (state variance 1 · observation variance 9) · gray band is missing · ※ synthetic data



02

PART 2 · Extensions

Smoothing and Nonlinear Filters

Information cutoff · EKF · UKF · particle filter

Filtering, smoothing and forecasting use different information sets

The information cutoff matters more than the name

Task	Information	Purpose
Filtering	Up to that time	Real-time state estimation
Smoothing	Up to a fixed observation cutoff	Re-estimate states before that cutoff
Forecasting	Up to now	Estimate future states

When it leaks

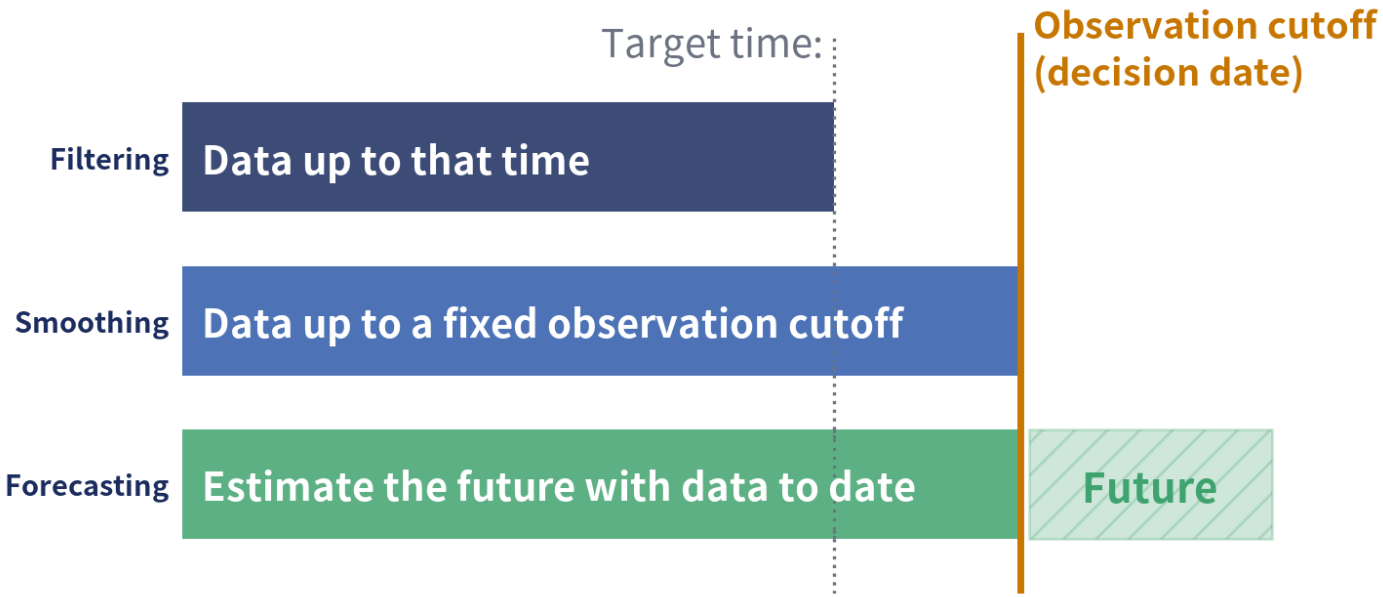
- Using smoothing built on the full sample as a past real-time signal

When it does not leak

- Smoothing last month's state with data up to today for today's decision

Smoothing is usable if the observation cutoff is respected

Comparing the three tasks by information set — this distinction returns in the Week 13 Bank of Korea case



Missing data

If all are missing, predict only; if some are, update with the observed part

Irregular spacing

Match F and Q to the actual time interval

A nonlinear model cannot be fully captured by mean and covariance

When the state and observation equations pass through functions f and h, linear-Gaussian computation is no longer exact

$$\underbrace{x_t = f(x_{t-1}) + w_t}$$

Nonlinear state equation

$$\underbrace{y_t = h(x_t) + v_t}$$

Nonlinear observation equation

Options

- EKF: straighten the curve around the current estimate
- UKF: pass representative points through the curve
- Particle filter: weight many candidate states

Perspective

- The particle filter is not a 'better Kalman' but a separate numerical method for general nonlinear state spaces

The four methods differ in what they approximate and what they cost

Look at strengths and limits together

Method	Computational intuition	Strengths	Limits
KF	Mean and covariance of a linear Gaussian model	Exact under those conditions	Limited under strong nonlinearity or non-Gaussianity
EKF	Linearize around the current estimate	Convenient for differentiable models	Linearization error · needs derivatives
UKF	Pass representative points through the curve	No Jacobian needed	Finite sigma points · moment approximation
Particle filter	Weight many candidate states	Approximates multimodal, non-Gaussian cases	Particle degeneracy · cost in high dimensions

The EKF borrows the KF computation via the local slope

Jacobians of the state and observation equations are evaluated at each period's estimate

$$F_t \approx \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_{t-1|t-1}}, \qquad H_t \approx \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_{t|t-1}}$$

How it works

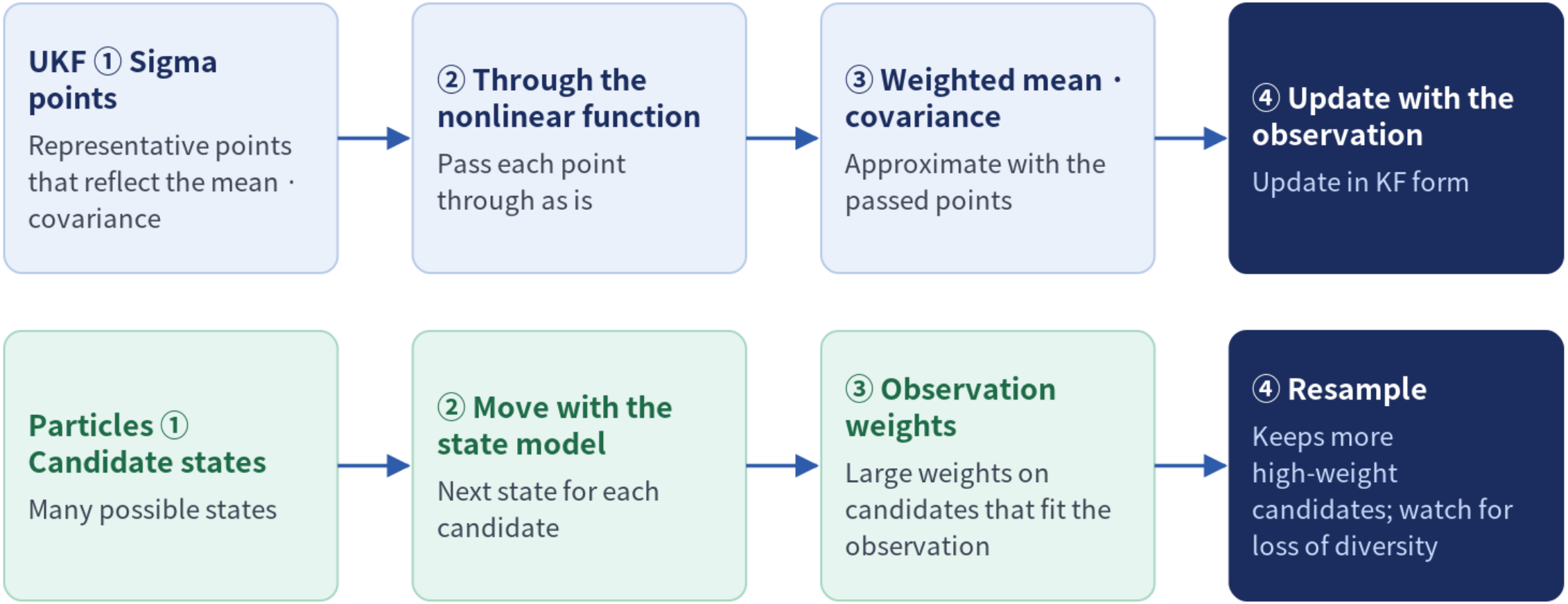
- Approximate the nonlinear function by a line around the current estimate
- Put those slopes in place of F and H and compute in KF form

Risk

- With a poor initial value or strong nonlinearity, it tracks a wrong approximation

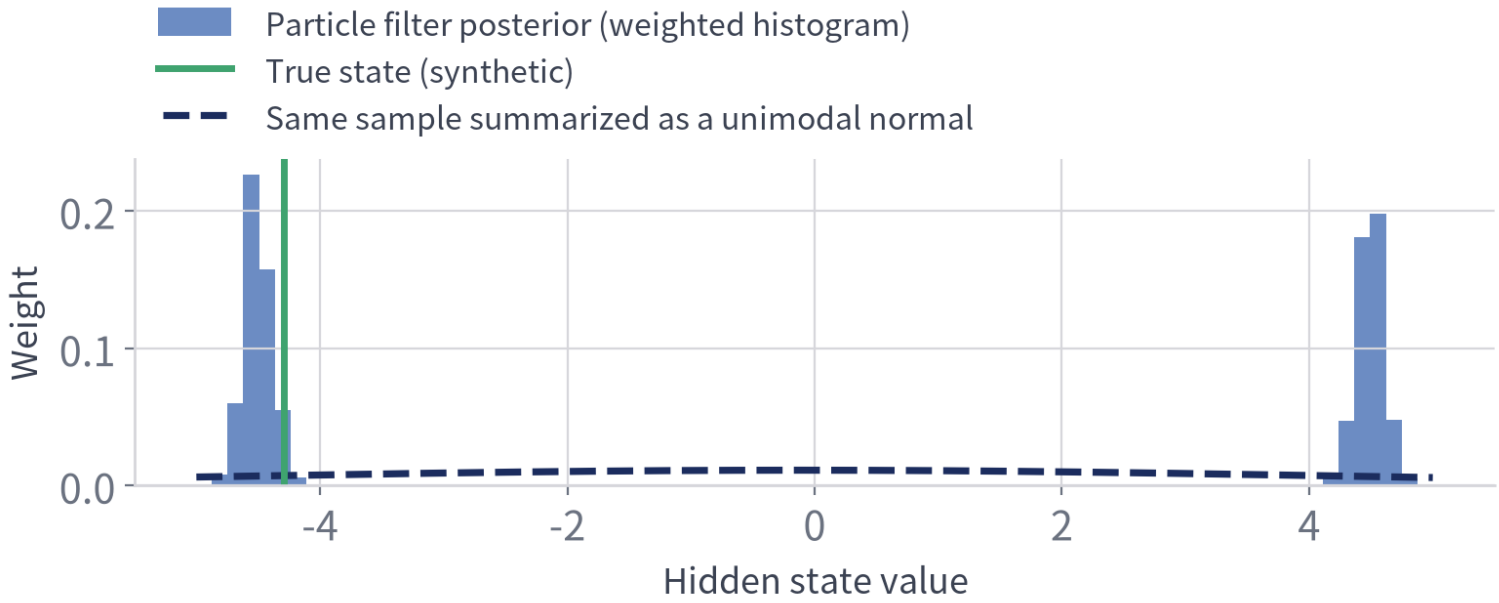
The UKF passes representative points; the particle filter, many candidates

Sigma points are not arbitrary: they are chosen to reflect the specified mean and covariance



With a squared observation, both positive and negative states survive

Observation = state squared + noise · the same observation can come from a positive or a negative state · ※ synthetic data



Mean-centered approximation

EKF·UKF summarize around a single-mode mean

Evaluation criteria

Check possible modes and intervals, not just a single mean error

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PART 3 · Applications

Dynamic Market Beta and Diagnostics

Dynamic regression · rolling OLS comparison · lab

Dynamic beta is linear in the state, so the KF is enough

Given the observed market return, the observation equation is linear in alpha and beta

$$\underbrace{r_{i,t} = \alpha_t + \beta_t r_{m,t} + v_t}_{\text{Observation equation: asset return}}$$

$$\underbrace{\begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix} = \begin{bmatrix} \alpha_{t-1} \\ \beta_{t-1} \end{bmatrix} + w_t}_{\text{State equation: random walk of alpha · beta}}$$

Benchmarks

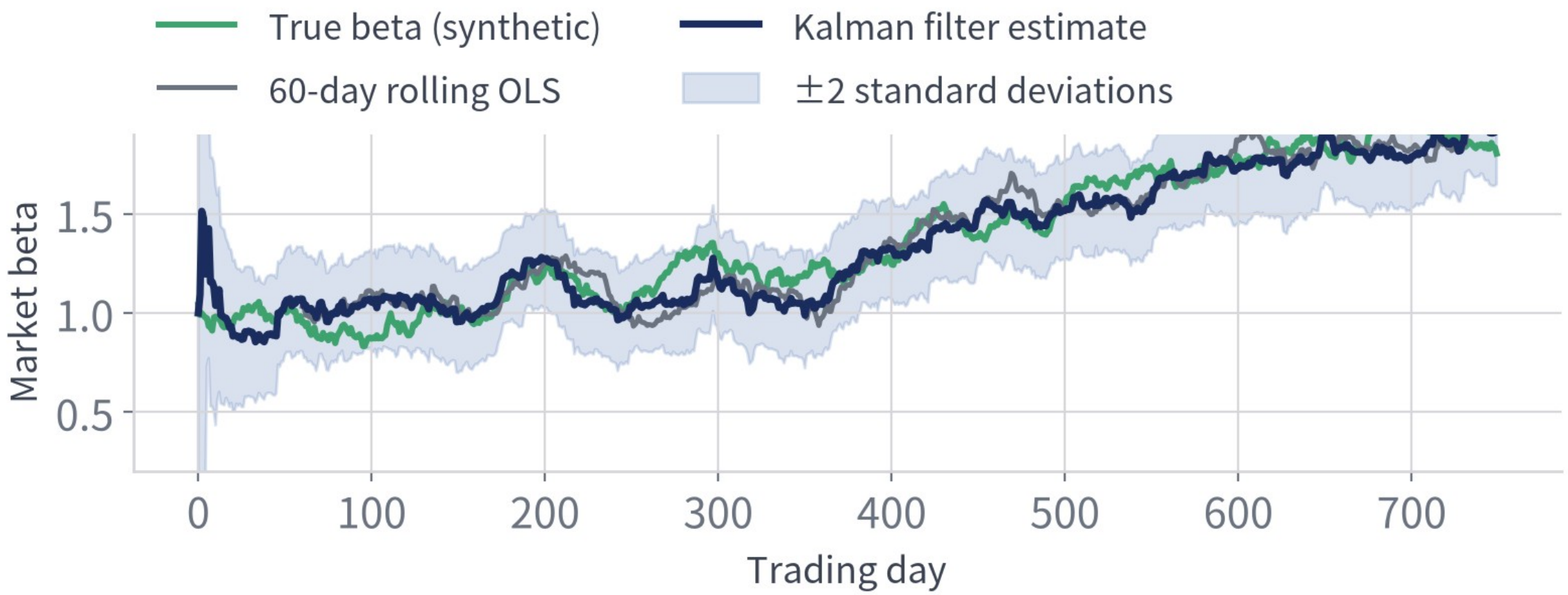
- Static OLS · rolling OLS
- Compare on out-of-sample innovations, hedge error, trading costs

Caution

- A dynamic regression does not require an EKF
- Dynamic beta does not automatically ensure cash neutrality or profitability

The Kalman beta gives an interval along with the estimate

Synthetic data with a random-walk true beta · compared with 60-day rolling OLS · ※ synthetic data



The lab runs from reproducing the hand calculation to dynamic beta

Paste into Claude Code, then check the resulting figures and tables

[Input] Prompt · Claude Code

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Write and run the Week 10 Kalman filter lab.
1. Reproduce the hand calculation with initial value 100,
   initial variance 4, Q 1, R 9, observation 106.
2. For the synthetic linear model, use the KF as the reference
   and compare the true state with the estimated interval.
3. Insert missing data and mark the prediction-only periods.
4. For the nonlinear squared-observation example, compare
   at least two of EKF, UKF and the particle filter.
   Do not force squared observations into a linear-observation KF.
5. Record the initial state, noise, seed and number of particles.
6. Separate experiments with known parameters
   from experiments with estimated parameters.
7. For the real financial application, estimate dynamic beta
   from past data only and compare it with rolling OLS.
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What to check

- Innovation autocorrelation
- Interval coverage rate
- Sensitivity to initial values
- Filter intervals are uncertainty under the specified model and parameters

A hidden continuous state is estimated by prediction and updating

Next week: the HMM, where the hidden state takes a few discrete categories

1 State space = state equation + observation equation

F · H · Q · R set how the state moves, how it is observed, and the size of the two noises.

2 Innovation and gain

The update is 'prediction + gain × innovation', and the gain is set by the relative size of uncertainties.

3 Information cutoff and model choice

Smoothing is usable if the cutoff is respected; for nonlinear or multimodal cases, move to EKF · UKF · particle filter.

Week 10 continuous state → Week 11 discrete regimes (HMM) → Kalman engine of Week 13 nowcasting